

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

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Code of Conduct:

I NS BHAVESH/192565086 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

NS BHAVESH

Signature of the student:

Annexure A

CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3. Given the concept map:

DataLink → MACAddress → LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→ DataLinkLayer → ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

SIMATS ENGINEERING

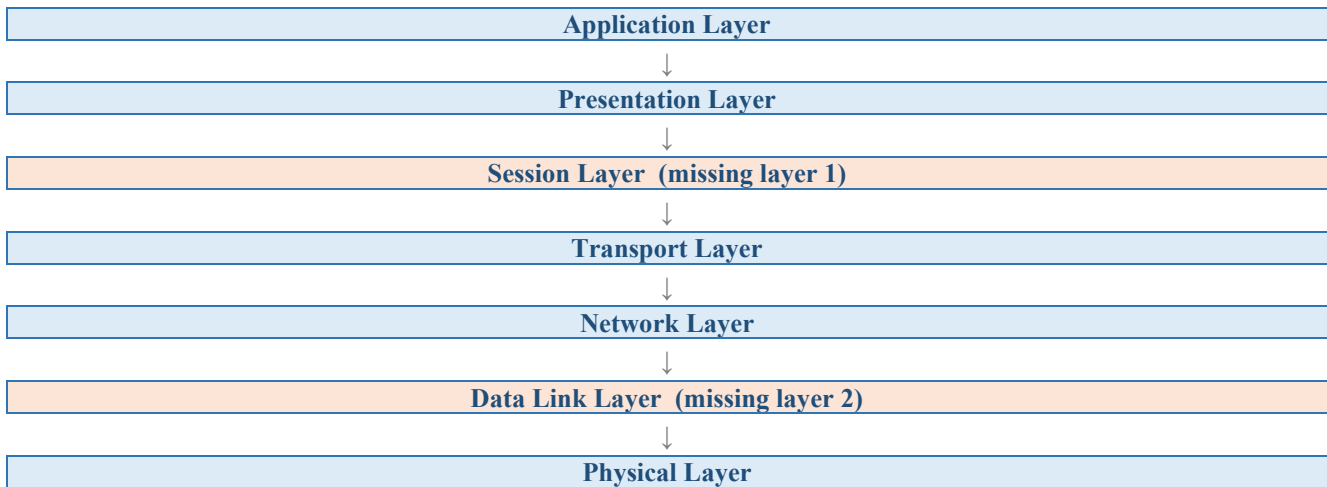
ASSESSMENT TOOL 2 — CONCEPT MAPPING

Course Name:	Computer Network
Course Code:	CSA0713
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Register Number:	192565086

Question 1

A partially completed concept map is given below: Application → Presentation → _____ → Transport → Network → _____ → Physical. Analyze the missing layers and explain how their functions are essential for reliable communication.

Completed Concept Map:



Analysis:

The two missing layers are the Session Layer (between Presentation and Transport) and the Data Link Layer (between Network and Physical).

The Session Layer establishes, manages, and terminates the communication session (dialogue) between two applications. It provides synchronization checkpoints and dialogue control (half-duplex/full-duplex), which allow an interrupted session to resume from the last checkpoint instead of restarting — this is essential for reliable, long-running communication such as file transfers or video calls.

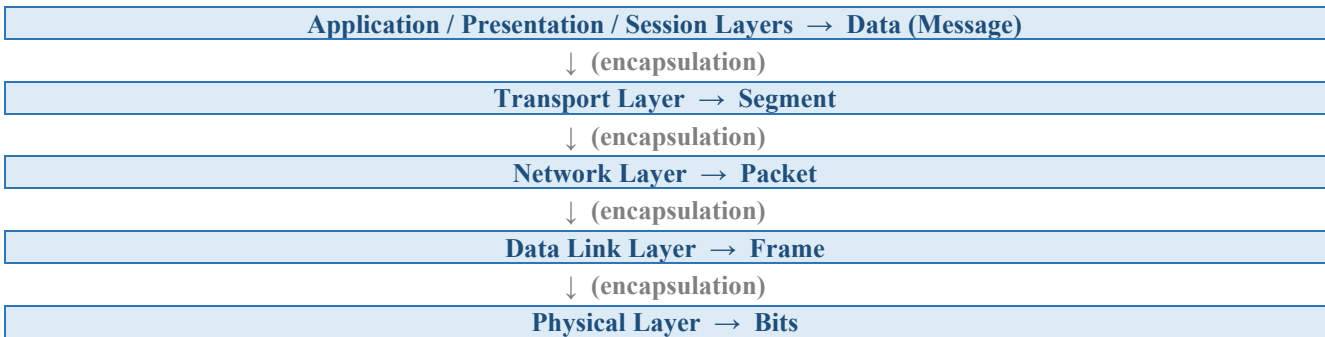
The Data Link Layer takes packets from the Network Layer and frames them with physical (MAC) addressing, plus error-detection fields such as a CRC. It is responsible for reliable node-to-node delivery across a single physical link, detecting bit errors introduced by the transmission medium and controlling access to that shared medium.

Without the Session Layer, applications would have no structured way to manage or recover an ongoing dialogue; without the Data Link Layer, packets could not be reliably placed on and taken off the physical medium, since there would be no local addressing or error-detection at the link level. Both layers are therefore essential links in the chain that makes end-to-end communication reliable.

Question 2

Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

Concept Map:



Analysis:

As user data moves down the OSI stack, each layer wraps (encapsulates) the data unit received from the layer above with its own header (and sometimes trailer), producing a new, layer-specific data unit: Data → Segment → Packet → Frame → Bits.

The Transport Layer converts application data into segments, adding port numbers and sequencing information needed for end-to-end delivery and reliability. The Network Layer wraps each segment into a packet, adding logical (IP) addressing needed for routing across networks.

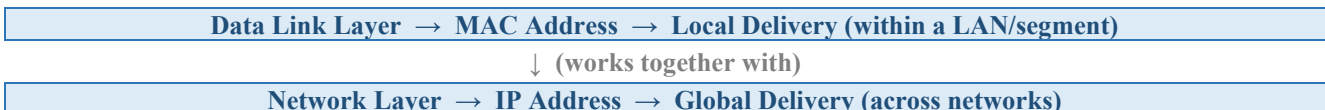
The Data Link Layer wraps each packet into a frame, adding physical (MAC) addressing and error-detection fields needed for reliable delivery across a single link, and finally the Physical Layer converts the frame into a raw stream of bits (electrical, optical, or radio signals) for transmission over the medium.

This progressive transformation supports transmission because each layer only needs to understand its own data unit and header — it does not need to know the internal structure of the layers above. On the receiving side, the reverse process (decapsulation) strips each header in turn, reconstructing the original data, which keeps the design modular, scalable, and easy to troubleshoot.

Question 3

Given the concept map: DataLink → MAC Address → Local Delivery; Network → IP Address → Global Delivery. Explain how these relationships ensure complete data delivery across networks.

Concept Map:



Analysis:

The Data Link Layer uses the MAC address, a physical address burnt into the network interface card, to deliver frames to the correct device within the same local network segment. MAC addressing is only meaningful on the local link and is not used for delivery beyond a router.

The Network Layer uses the IP address, a logical address, to identify the source and destination across different networks and to make routing decisions so a packet can travel across multiple intermediate networks (via routers) to reach a distant destination — this provides global delivery.

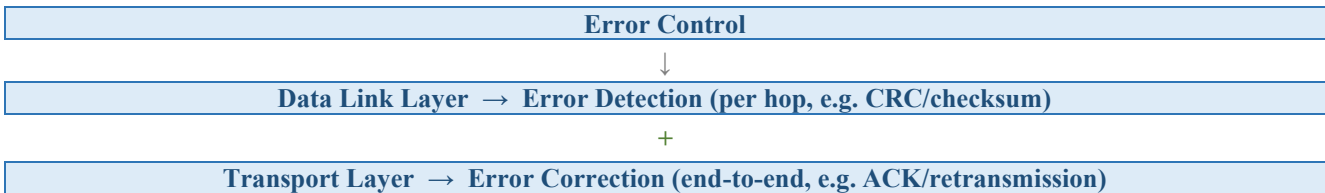
These two addressing schemes work together through protocols such as ARP, which maps an IP address to the correct MAC address at each hop: a router forwards a packet based on the destination IP address, but on each individual link it must re-encapsulate the packet in a frame addressed to the MAC address of the next hop.

Together, MAC addressing provides reliable last-hop, local delivery while IP addressing provides the end-to-end path across the internetwork; without local (MAC) delivery a packet could never reach the next router or the final host, and without global (IP) addressing/routing, a packet could never leave its own local network — the two layers must cooperate to achieve complete data delivery.

Question 4

A concept map shows: Error Control → Data Link Layer → Error Detection; → Transport Layer → Error Correction. Analyze how the interaction between these layers improves reliability in data transmission.

Concept Map:



Analysis:

The Data Link Layer performs error detection on each individual physical link using techniques such as CRC or checksums. If a frame is found to be corrupted, it is typically discarded (or, in some link protocols, retransmitted locally), catching errors introduced by noise or interference on that specific hop before they propagate further.

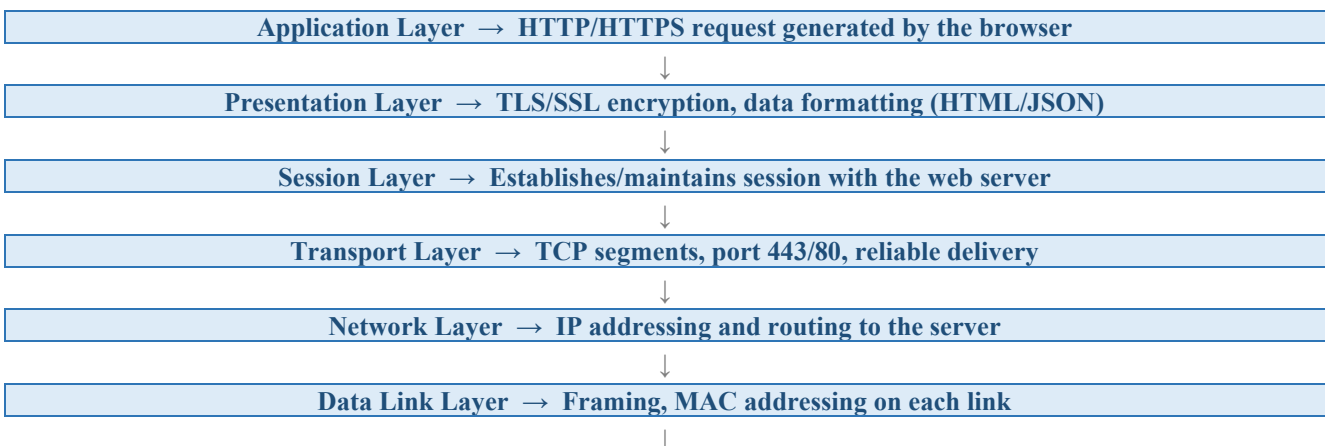
The Transport Layer performs error correction on an end-to-end basis, independent of how many links or routers the data passed through. Using sequence numbers, acknowledgements, and timers, it detects missing or corrupted segments anywhere along the path and requests retransmission from the original sender, guaranteeing the receiving application ultimately gets a complete, correct copy of the data.

The interaction between the two improves overall reliability because they defend against errors at two different scopes: the Data Link Layer prevents locally-corrupted frames from being treated as valid, reducing wasted processing further up the stack, while the Transport Layer provides a safety net that catches any errors or losses that slip through multiple hops (including packets dropped by routers due to congestion, not just bit errors) — together giving layered, defense-in-depth reliability rather than relying on a single point of error control.

Question 5

Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

Concept Map (Web Browsing, e.g. HTTPS request in a browser):



Analysis:

Each layer depends on the correct functioning of the layer below it, so a failure at any single layer breaks the entire browsing session, not just that layer's own function.

If the Physical Layer fails (e.g., cable unplugged or no Wi-Fi signal), no bits can be transmitted at all, so nothing above it can function. If the Data Link Layer fails (e.g., a corrupted frame or MAC conflict), the packet never reaches the next hop, so the Network Layer cannot route it further.

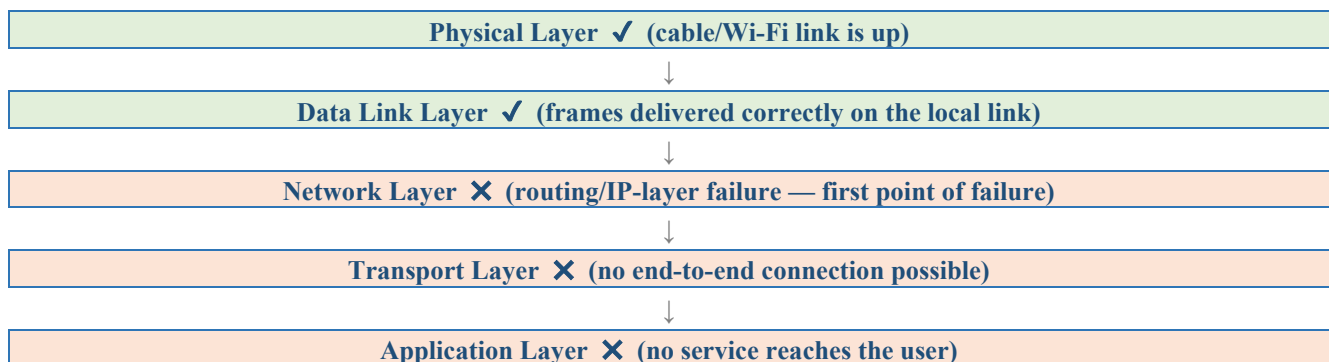
If the Network Layer fails (e.g., no route to the server's IP, or an incorrect default gateway), the request cannot cross into the wider internet even though the local link works, so the Transport Layer's TCP connection can never be established. If the Transport Layer fails (e.g., the server refuses the TCP connection or packets are lost without recovery), no reliable byte stream reaches the Application Layer.

If the Presentation Layer fails (e.g., a TLS handshake/certificate error), the browser may refuse to render the page even though data physically arrived, and if the Application Layer itself fails (e.g., the web server returns an error or the browser cannot parse the HTTP response), the user sees no usable page despite every lower layer working correctly. This dependency chain is why web browsing (and any OSI-based communication) requires every layer to function correctly, end to end.

Question 6

A network issue is represented in the concept map: Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗. Analyze the problem and explain how the concept map helps in troubleshooting.

Concept Map (Fault Isolation):



Analysis:

Since the Physical and Data Link layers are both working, the physical medium is intact and the device can successfully send and receive frames within its local network segment — so the fault is not a cabling, NIC, or local-link problem.

The failure first appears at the Network Layer, which strongly suggests an IP-layer problem such as a missing or incorrect default gateway, an incorrect IP address/subnet mask, a routing table misconfiguration, or a router/DNS failure that prevents packets from being routed beyond the local network.

Because the Transport and Application layers depend entirely on the Network Layer to deliver packets end-to-end, their failure is a direct downstream consequence of the Network Layer fault rather than a separate, independent problem — fixing the Network Layer issue (e.g., correcting the gateway/routing configuration) would very likely restore Transport and Application functionality as well.

This is exactly why the concept map is useful for troubleshooting: by checking each OSI layer from the bottom up, an engineer can pinpoint the lowest layer at which failure first occurs (here, the Network Layer) and know that all layers above it will also fail as a consequence. This bottom-up, layer-by-layer method avoids wasting time investigating the Application Layer configuration when the real fault lies in IP routing, making diagnosis faster and more systematic.